

Higher frequencies of CD161+ circulating T lymphocytes in allergic rhinitis patients compared to healthy donors.

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BACKGROUND: Th17 is a subset of T-helper lymphocytes that produce proinflammatory cytokines, mainly IL-17. Serum IL-17 is increased in allergic patients and relates to clinical severity. Recently, it has been reported that CD161 is a highly upregulated gene in Th17 clones and all IL-17-producing cells are contained in CD161(+) T cells. This study aimed at comparing the frequency of peripheral CD161(+) T cells in patients with allergic rhinitis (AR) and in healthy controls and at relating CD161 expression with symptom severity. METHODS: Forty-four patients with AR and 29 healthy non-allergic subjects were evaluated. CD161 expression was evaluated on CD3(+), CD4(+) and CD8(+) cells by double immunofluorescence staining and fluorescence activated cell sorter analysis. Symptom severity was assessed by the Visual Analogue Scale. RESULTS: Allergic patients showed a significantly higher frequency of CD3(+)CD161(+), CD4(+)CD161(+) and CD8(+)CD161(+) cells than healthy non-allergic subjects ($p < 0.0001$). Moreover, the expression of CD161 cells was significantly related to clinical severity. CONCLUSIONS: This study provides evidence that a higher frequency of CD161(+) T cells is present in the peripheral blood of AR patients. Copyright © 2012 S. Karger AG, Basel.